

Interactive effects of dispositional mindfulness and PETTLEP imagery training on basketball shooting performance: A randomized controlled trial

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ABSTRACT

The purpose of this study was to examine the interactive effects of dispositional mindfulness and visualized PETTLEP imagery training on basketball mid-range shooting performance and retention. Seventy-three participants (M age = 20.32 ± 1.09) with high/low dispositional mindfulness (high $n = 35$; low $n = 38$) selected out of 302 college students were randomly assigned into the following six groups: (a) high mindfulness internal imagery (H-II, $n = 13$); (b) high mindfulness external imagery (H-EI, $n = 11$); (c) high mindfulness control (H-CO, $n = 11$); (d) low mindfulness internal imagery (L-II, $n = 13$); (e) low mindfulness external imagery (L-EI, $n = 12$); and (f) low mindfulness control (L-CO, $n = 13$). Participants engaged in a pretest to measure their basketball shooting performance, then participated in a 6-week (3 times/per-week) intervention, plus a posttest and retention test. A three-way 2 (high/low mindfulness) $\times$ 3 (treatments: internal-, external imagery, and control) $\times$ 3 (measurement time: pretest, posttest, and retention) mixed ANOVA statistical analysis found dispositional mindfulness interacted with treatments and measurement time. The main effects showed high dispositional mindfulness performed better than low dispositional mindfulness, and internal imagery training performed better than external imagery training on mid-range basketball performance at retention. The 3-way interaction indicated that when using either internal or external imagery, high dispositional mindfulness performed better than low mindfulness on retention but not posttest. For 2-way interaction, high dispositional mindfulness performed better than low dispositional mindfulness on retention but not posttest. Our results extended current knowledge on sport imagery and dispositional mindfulness and gained several theoretical implications for researchers. The limitations, future research directions, and practical implications were also discussed.

Mindfulness, defined as “paying attention in a particular way: on purpose, in the present moment and non-judgmentally” (Kabat-Zinn, 1994, p. 4), has received much attention from researchers. Although there are many definitions of mindfulness (Khouri et al., 2017), Kabat-Zinn’s (1994) definition captures the way mindfulness is practiced and characterized. Specifically, this definition highlights that mindfulness is a state of consciousness that involves the cognitive

process of attention and awareness. It indicates that mindfulness is not an automatic experience; rather, one must purposely bring their attention to the present in a non-judgemental manner. Further, Kabat-Zinn’s (1994) definition outlines the basic axioms of mindfulness such as (a) intention-that is, one intends to practice mindfulness with several sub-intentions (i.e., self-regulation, self-exploration, and self-liberation); (b) attention-one has to focus on their inner and external

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experiences from moment to moment; and (c) attitude—one has to attend to consciousness and awareness in a non-judgmental, non-interpretative, accepting, and open manner (Shapiro et al., 2006).

Along with the evolution of the research, mindfulness has been conceptualized/studied as either a state (i.e., a momentary condition) or a trait (i.e., a stable characteristic). The state mindfulness is characterized as having clear awareness, non-conceptual, non-discriminatory awareness, flexible attention and awareness, empirical stance toward reality, present-oriented consciousness, and stable attention and awareness (Brown et al., 2007). In contrast, trait mindfulness (known as dispositional mindfulness, DM), is defined as “... tendency for having an open or receptive awareness and attention, which is reflected in a more regular or sustained consciousness of ongoing events and experiences” (Brown & Ryan, 2003, pp. 822–823). Research found through formal and informal mindfulness practice/intervention one can cultivate dispositional mindfulness (Carmody et al., 2009), and experience several cognitive attributes such as pure attention, experiential acceptance, values clarification, self-regulation, clarity, exposure, flexibility, non-attachment, and less rumination; which are beneficial to psychological adjustment (Birrer et al., 2012).

Because dispositional mindfulness is closely linked to one's cognitive state of attention awareness, researchers have examined how dispositional mindfulness influences motor control and performance. For example, in a study that examined how a brief mindfulness induction influences motor balance, Kee et al. (2012) recruited 32 university students who engaged in a 6-min mindfulness intervention and performed a balance task. They found no main effects of the mindfulness intervention on the balance task. However, dispositional mindfulness interacted with mindfulness induction; specifically, in the intervention group, those with high dispositional mindfulness performed better than low dispositional mindfulness.

In the same manner, Chiu et al. (2019) hypothesized that those with high dispositional mindfulness can maintain attention longer and not be disturbed when using unrelated self-talk in motor tasks. They assigned 49 college students (high mindfulness = 23; low mindfulness = 26) to engage in instructional self-talk or unrelated self-talk and perform line tracking and standing long-jump tasks. As expected, dispositional mindfulness interacted with self-talk in line tracking. Specifically, low dispositional mindfulness performed worse than high dispositional mindfulness in line tracking when using unrelated self-talk. Chiu et al. (2019) explained that line tracking requires performers to maintain attention longer. If low mindfulness performers use unrelated self-talk, performance deteriorates.

Recently, Liu et al. (2022) argued that past research combining dispositional mindfulness and psychological interventions (i.e., Chiu et al., 2019; Kee et al., 2012) was conducted in the laboratory with no audience. They hypothesized that if participants performed motor tasks under the presence of others they would have different results. They recruited 46 male participants (high dispositional mindfulness = 23; low dispositional mindfulness = 23) who performed dart-throwing and two-hand coordination under pressure (with the audience) and non-pressure (performed alone) using instructional self-talk and unrelated self-talk. A 3-way interaction found those with low mindfulness performed worse in two-hand coordination under pressure when using unrelated self-talk. Liu et al. (2022) explained that low dispositional mindfulness is a disadvantage in maintaining attention when performing a continuous motor task (i.e., two-hand coordination) under pressure and using unrelated self-talk.

Past research that combines dispositional mindfulness and psychological interventions in examining motor performance has theoretical significance. Theoretically, dispositional mindfulness is an individual's cognitive tendency to perceive environmental stimuli and maintain attention to the task at hand (Brown & Ryan, 2003), so those with high dispositional mindfulness perform better than those with low dispositional mindfulness in certain motor tasks. From a motor learning and control perspective, maintaining attention is important for motor skill

performance. To successfully accomplish a motor task, performers need to focus attention on required components such as the standing position, location and angle of equipment, distance, size and direction of target objects, and timing for motor execution (Schmidt & Lee, 2020). For example, a tennis player who wants to successfully return an opponent's serve must carefully look at the incoming ball's bouncing height and direction, focus on the racket's swing pathway and choose the right timing to hit the ball. Attention has been defined as “... a behavioral and cognitive process that ignores environmental unrelated stimuli and selects a related cue, maintaining such focus to complete a given task” (Anderson, 2004, p.519). A tennis player can successfully return a fast serve if he/she can cognitively ignore irrelevant stimuli in the environment (e.g., audience, umpire), and focus attention on the bouncing height, direction, and speed of the incoming ball to make a good return. Schmidt et al. (2019) contended that attention allows performers to efficiently allocate cognitive resources and deal with related information efficiently to successfully complete a motor task.

Thus, this study examined how dispositional mindfulness combined with motor imagery in influencing sport performance. Motor imagery is a widely used psychological skill that requires performers to simulate sports experiences and integrate all sensations (i.e., visual, olfactory, auditory, kinesthetic, gustatory) to create or recreate a cognitive representation without overt behavior (Jeannerod, 1994; Vealey & Forlenza, 2015). The effects of motor imagery on performance have been examined in many contexts such as rehabilitation (Multhaupt & Beuth, 2018), art performance (Pavlik & Nordin-Bates, 2016), medicine (Slimani et al., 2016), dance (Nordin & Cumming, 2005), and exercise (Kossert & Munroe-Chandler, 2007). Researchers also examined the effectiveness of motor imagery on athletic performance in sports such as biathlon (Gros Lambert et al., 2003), golf (Smith et al., 2008), gymnastics (Battaglia et al., 2014), tennis (Dana & Gozalzadeh, 2017), basketball (Lu et al., 2020), and table-tennis (Caliari, 2008).

In this study, we examined the combination of dispositional mindfulness and motor imagery in basketball mid-range shooting. Specifically, we applied Lu et al.'s (2020) visualized PETTLEP imagery approach (i.e., using participants' successful performance videos as an intervention tool) to conduct our study for several reasons. First, PETTLEP imagery training is a sound theoretical model of imagery training developed by Holmes and Collins (2001). Holmes and Collins (2001) used the functional equivalence hypothesis and contended that imagery shares the same neuron networks in the brain as actual movements. If one engages in imagery activity, the neurophysiological response areas in the brain are similar to a real movement. Further, to gain the best effects of imagery, Holmes and Collins (2001) proposed that imagers should integrate 7 fundamental elements of imagery –P (physical), E (environment), T(task), T(timing), L(learning), E(motion), and P (perspective) into the training program. The PETTLEP imagery training has been found effective in enhancing motor performance (e.g., Battaglia et al., 2014; Buck et al., 2016; Norouzi et al., 2019).

Despite its effectiveness, Lu et al. (2020) argued that past imagery training generally uses “imagery scripts” to do imagery. Lu et al. (2020) argue that imagers who are dyslexic or have a hearing problem would have an “imagery gap” during imagery because they can't produce an image in mind. According to imagery theory, if one intends to do imagery, he/she needs to recognize, interpret, and create images in mind (Kosslyn, 2005). Participants with dyslexia or hearing problems asked to read/hear an “imagery script” would have difficulty. Thus, prior to imagery intervention, Lu et al. (2020) filmed participants' own successful 3-point basketball shot then asked participants to watch the video, and imagine it before a real practice. They found experimental groups performed better than the control group after 8-week visualized PETTLEP imagery training.

Thus, we intended to use this newly developed visualized PETTLEP imagery approach to conduct our study. Further, past research that combined dispositional mindfulness and psychological interventions (i.e., Chiu et al., 2019; Kee et al., 2012; Liu et al., 2022) generally

examined immediate effects but did not consider the effects on retention. Retention is the process of transferring new information into long-term memory; it is the motor memory or the persistence of motor performance (Schmidt et al., 2019). Thus, we intended to combine visualized PETTLEP imagery training and dispositional mindfulness on basketball mid-range shooting performance and retention. By doing so, we expect to fill the gap in the research. Further, when applying PETTLEP imagery, the last acronym "P" represents the "perspective" of the imagery training. Some researchers suggested that for closed motor skills, such as tennis serve, internal perspective is more effective (Dana & Gozalzadeh, 2017). Accordingly, whether internal imagery will lead to better performance than external imagery was one of our research questions.

In sum, the purpose of the present study was to examine the interactive effects of dispositional mindfulness and visualized PETTLEP imagery training on basketball mid-range shooting performance and retention. We hypothesized that dispositional mindfulness will serve as a moderator (Hayes, 2017) and interact with visualized PETTLEP imagery in mid-range shooting performance and retention. Further, we hypothesized that internal PETTLEP imagery is more effective than the external

perspective on basketball mid-range shooting performance and retention. Moreover, we hypothesized that high mindfulness participants will have better performance than low mindfulness participants on basketball mid-range shooting performance and retention.

1. Method

1.1. Participants

After obtaining ethical approval from a local institutional ethics committee (TSMH IRB No. 22-008-B), we visited a physical education office of a local university at the beginning of the semester. At a regular administration meeting, we described our project, purposes, procedures, potential benefits/risks, and rights of the participants to the director and physical education (PE) teachers. After the director and PE teachers understood our project, they agreed to let us recruit students from the PE classes as participants. In the first PE class of the semester, we briefed students on the project. Those who agreed to participate signed the consent form, and completed a demographic questionnaire and Chinese version of the Mindful Attention Awareness Scale (CMAAS; Chang et al.,

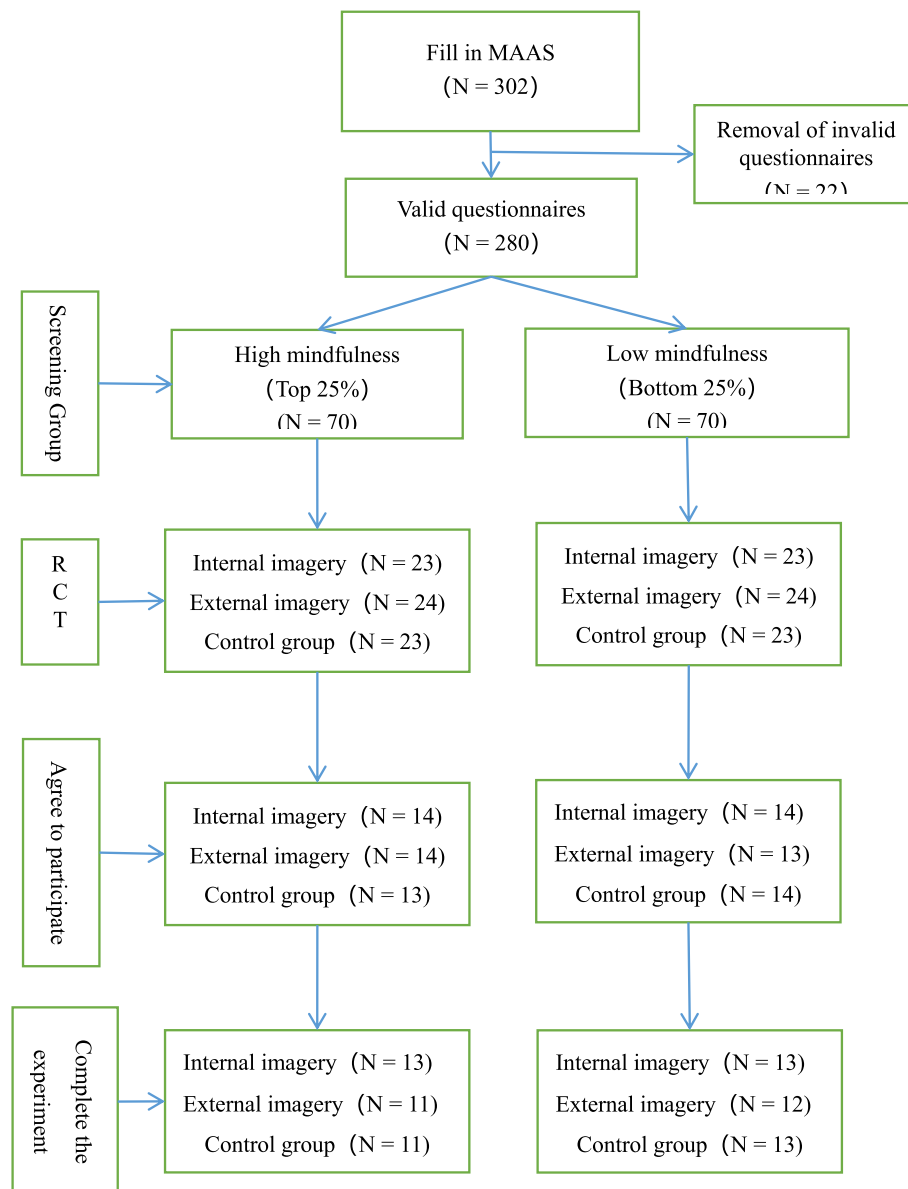


Fig. 1. Research process flowchart.

2011). We collected valid data from 302 participants and identified the upper 25% and lower 25% as high vs. low dispositional mindfulness. A total of 82 students (high mindfulness = 41; low mindfulness = 41) were recruited, but 9 participants declined before the experiments began. The final sample size was 73 (M age = 20.32 yrs, SD = 1.09; females = 13, males = 60). We randomly assigned them to 6 groups: 13 participants in the high mindfulness internal imagery group (H-II), 11 participants in the high mindfulness external imagery group (H-EI), 11 participants in the high mindfulness control group (H-CO), 13 participants in low mindfulness internal imagery group (L-II), 12 participants in low mindfulness external imagery group (L-EI), and 13 participants in the low mindfulness control group (L-CO; see Figure 1). All participants were ordinary college students. They participate in the college general PE classes as one of their requirements. They have previous basketball experience in elementary and middle schools but did not participate on basketball teams. Further, they have never participated in any psychological skills training (PST).

1.2. Measures

We used the following measures for our study.

1.2.1. Demographic questionnaire

A demographic questionnaire was used to collect participants' basic information including gender, age, weight, height, department, physical condition, years of experience in basketball, and a 4-digit ID for tracking.

1.2.2. Mindful Attention Awareness Scale (MAAS)

We used the Chinese version of the Mindful Attention Awareness Scale (CMAAS; Chang et al., 2011) to assess participants' dispositional mindfulness. The MAAS (Brown & Ryan, 2003) is a 15-item trait measure of dispositional mindfulness. A sample question is "I tend not to notice feelings of physical tension or discomfort until they grab my attention." Items are rated on a 6-point scale, ranging from 1 (almost always) to 6 (almost never). Total scores are calculated by summing all 15 items with higher scores indicating higher levels of dispositional mindfulness. Chang et al. (2011) reported that CMAAS had adequate internal reliability of 0.88, and 0.90 in a two times measurement.

1.2.3. Movement Imagery Questionnaire-revised (MIQ-R)

We used the revised Chinese Version of the Movement Imagery Questionnaire (MIQ-R; Lin, 2011), which was revised from the 8-item MIQ (Hall & Martin, 1997), to assess participants' imagery ability. The MIQ-R has 4 items for measuring kinesthetic imagery ability and 4 items for measuring visual imagery ability. To assess imagery ability, participants read the description of 4 movements (i.e., raising the leg, vertical jump, moving a non-dominant hand, and bending the body); then they feel and imagine the action, and finally rate the ease or difficulty of imaging the movement on a 7-point Likert scale ranging from 1 (very difficult to feel/see) to 7 (very easy to feel/see). Higher scores represent better imagery ability. Lin (2011) reported that the revised Chinese MIQ-R has adequate reliability illustrated by composite reliability (CR) –i.e., visual motor imagery = 0.88, and kinesthetic motor imagery = 0.82.

1.2.4. Manipulation check questionnaire

The manipulation check questionnaire was designed to confirm whether participants followed our experimental instructions. After the intervention, we asked the experimental groups to answer seven questions, using a 7-level Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree), to confirm that they complied with the intervention procedures (Marshall & Wright, 2016; Williams et al., 2013). Sample questions were as follows: "I was focused in practice"; "I followed the script during imagery practice"; and "the intervention worked for me."

1.3. Experimental procedures

We scheduled participants to arrive at the experimental basketball court located in a quiet corner of the gym one by one. On the first visit, each participant completed the Chinese version of the MIQ-R to assess their imagery abilities. Then, all participants, including all six groups, were asked to perform a mid-range shot from 5 spots at the angles of 0, 45, 90, 135, and 180 on the basketball court (see Figure 2). At each spot, they were asked to perform 5 shots. The performance of all 25 shots was used as the pretest data. To assess mid-range shot performance, we adopted the Wakefield and Smith, D. (2009) scoring system as follows: *touching neither the rim nor the basketball board* = 0 points; *hitting the basketball board first but no goal* = 1 point; *hitting the rim first of the basketball board but no goal* = 2 points; *hitting the basketball board first then goal* = 3 points; *hitting the rim of the basketball board first then goal* = 4 points; *directly shooting into the basket* = 5 points. Thus, the maximum score of the mid-range shot is 25 points, and the minimum score is 0 points. The distance of the mid-range shot is 460 cm from the basketball board as a previous study determined that was the optimal distance (Okazaki & Rodacki, 2012). According to International Basketball Federation (FIBA) rules, we used the No.7 ball for males and No.6 for females.

1.4. Intervention

Imagery Video Preparation: To make the videos, we filmed each participant's successful mid-range shot in the 5 shooting spots on the basketball court a week before the formal experiment. For internal imagery, the photographer stood directly behind the participant to get the first perspective images, including the release of the ball, the path of the ball, and the goal scored. To film external imagery, the photographer stood at the top of the triangle perpendicular to the shooting spots and the participant. We used these individualized successful videos for each experimental participant when they engaged in intervention.

Intervention Execution: First, we gave participants 5–10 min to warm up. Then, they practiced 2 shots from the 5 shooting points. Once they were ready, we started our intervention in the following steps: (1) participants viewed their successful videos that had already been made (the internal perspective video for the internal imagery group; the external perspective video for the external imagery group); (2) they were asked to view these videos then imagine it in their minds as they held the basketball (physical, P); stood in the basketball court (environment, E), ready to perform a mid-range shooting (task, T); performed it in real-time (time, T), emphasized learning this skill through practice (learning, P), experienced the real emotion during basketball shooting (emotion, E), and relived the experience of their success in video perspective (perspective, P); (3) participants performed the mid-range shot five times from 0 angles by receiving an assistant's pass; (4) repeat steps 1–2 to complete the other 4 shooting spots (i.e., 45, 90, 135, and 180°). To avoid fatigue, we gave participants 1 min of rest before proceeding to the next shooting spot (Lu et al., 2020). The control groups did the same procedure as the experimental groups without viewing videos and imagery, that is to say, they only completed physical practice including five shooting spots and five shots per spot.

1.5. General procedures

After the pretest, the experiment was conducted three times a week for six weeks (i.e., if they came on Monday, they came again on Wednesday, then, Friday; and if they came on Tuesday, they came again on Thursday, then, Saturday). The participants rested for 10 min after the final intervention, then, we conducted a posttest, which was identical to the pretest. For the control group, the posttest was conducted after they completed mid-range shot training, with the same content and procedure as the experimental groups. Further, the retention test was performed 1 week after the posttest (Lola & Tzetzis, 2021). The content

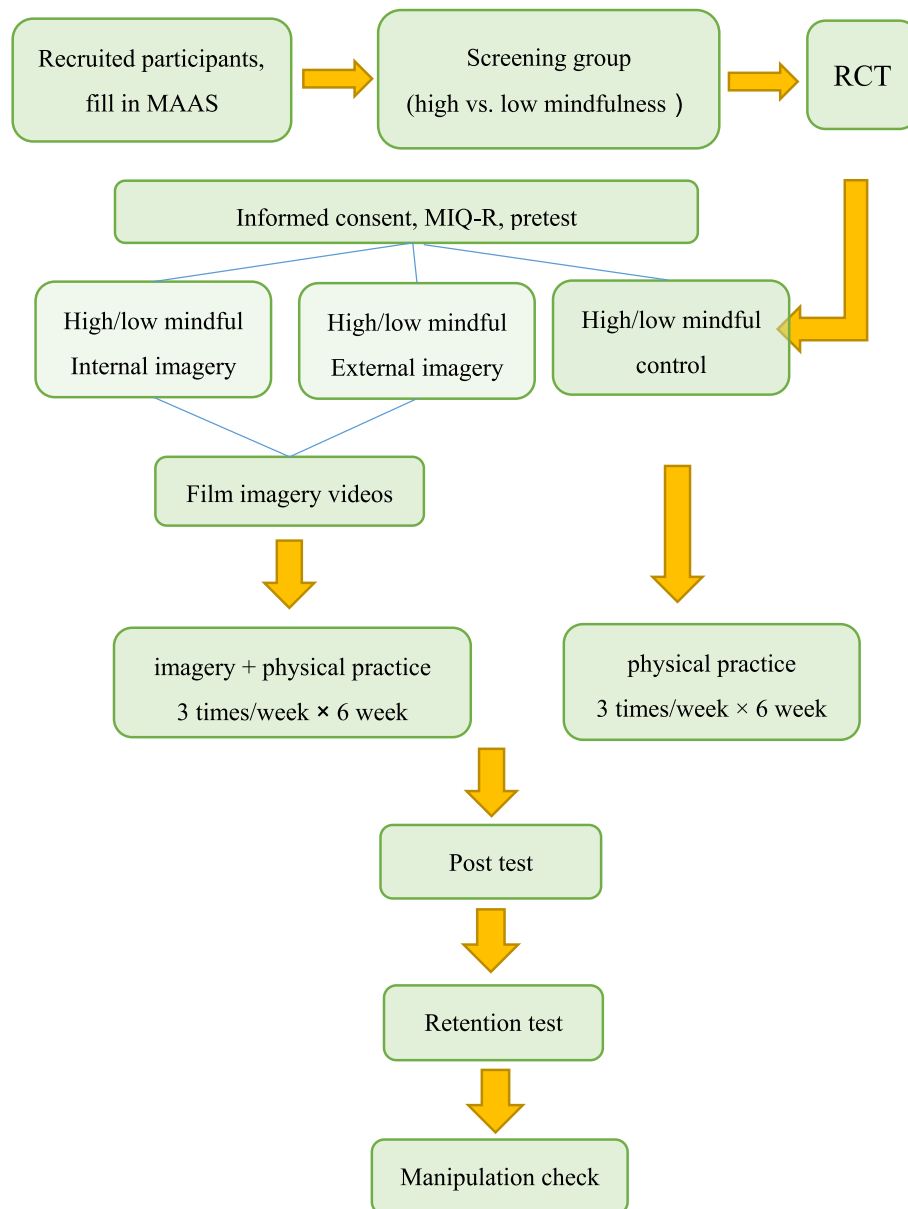


Fig. 2. The flowchart of experimental process.

and procedure of the retention test were identical to the posttest. The whole experiment process is shown in Figure 2.

1.6. Statistical analysis

We used a three-way 2 (high vs. low dispositional mindfulness) $\times$ 3 (treatments: internal imagery vs. external imagery vs. control) $\times$ 3 (measurement time: pretest vs. posttest vs. retention test) mixed ANOVA design to determine whether high vs. low mindfulness perform differently under different imagery and measurement time. Significant effects were followed-up with multiple comparisons with Box's Test for Equivalence of Covariance Matrices and Mauchly test to meet the sphericity assumption. We used repeated-measure analysis of variance to test the differences in performance within groups at different measurement times. The effect size was calculated and reported using partial η^2 values. The significance level was set at an alpha level of 0.05 for all statistical analyses.

2. Results

2.1. Preliminary analyses

Two-way ANOVA revealed that there are no significant differences between groups in age ($M = 19\text{--}20$ years, $SD = 0.75\text{--}1.36$), $F(2,71) = 1.347$, $p = .266$; height ($M = 173.75\text{--}178.77$ cm, $SD = 4.81\text{--}9.86$), $F(2,71) = 0.665$, $p = .517$; weight ($M = 64.33\text{--}70$ kg, $SD = 5.85\text{--}16.21$), $F(2,71) = 0.125$, $p = .882$; and year of experience in basketball, ($M = 13.0$ year, $SD = 4.04$), $F(2,71) = 1.179$, $p = .314$. We also analyzed participants' imagery abilities. The two-way ANOVA found no significant differences among groups. Further, all of the groups reported that they didn't receive any psychological interventions during experiments. Further, we analyzed manipulation check questionnaires with one-way ANOVA and found no significant differences among groups. Participants reported a mean score ranging from 6.08 to 6.54 for their involvement and effectiveness in visualized PETTLEP intervention; indicating that all participants displayed a high level of adherence to the intervention protocol.

2.2. Performance data

2.2.1. Main effects

There was a significant main effect of dispositional mindfulness on basketball mid-range shots, $F(1, 67) = 5.536$, $p = .022$, partial $\eta^2 = 0.076$ (low mindfulness $M = 60.57$, $SD = 1.84$; high mindfulness $M = 66.84$, $SD = 1.93$). There was no main effect of imagery training on basketball mid-range shots, $F(2, 67) = 2.280$, $p = .110$, partial $\eta^2 = 0.064$ (internal imagery $M = 66.92$, $SD = 2.23$; external imagery $M = 60.01$, $SD = 2.37$; control group $M = 64.19$, $SD = 2.33$). Furthermore, there was a significant difference among measurement time, $F(2, 66) = 127.57$, $p < .001$, partial $\eta^2 = 0.794$; (pretest $M = 59.72$, $SD = 1.46$; posttest $M = 68.32$, $SD = 1.36$; retention $M = 63.07$, $SD = 1.28$). Specifically, the posttest was better than the pretest and retention, and retention was better than the pretest (see Table 1).

2.2.2. Interaction effects

After main effect analyses, we used Box's Test for Equivalence of Covariance Matrices to compare variation in multivariate samples and found no significant differences (Box's $M = 37.84$). Also, we used Mauchly's sphericity test to validate a repeated measure analysis of variance (ANOVA) and found no significant difference. So, we examined 3-way and 2-way interactions among variables.

Table 1

A summary table of descriptive statistics of basketball mid-range shot of all study groups in 3-way and 2-way interaction.

Variables		Mid-range shot		
		Pretest	Posttest	Retention
Internal imagery	High mindfulness (n = 13)	M = 64.23 (SD = 9.67)	M = 75.54 (SD = 9.79)*	M = 74.69(SD = 9.65)
	Low mindfulness (n = 13)	M = 58.38 (SD = 12.63)	M = 68.00 (SD = 11.57)*	M = 60.69(SD = 10.90) #
External imagery	High mindfulness (n = 11)	M = 60.00 (SD = 18.74)	M = 64.36 (SD = 16.60)*	M = 64.00(SD = 15.39)
	Low mindfulness (n = 12)	M = 54.17 (SD = 8.90)	M = 64.00 (SD = 9.15)*	M = 53.50(SD = 7.50) #
Control group	High mindfulness (n = 11)	M = 62.64 (SD = 9.60)	M = 70.28 (SD = 8.63)*	M = 65.82(SD = 8.63)
	Low mindfulness (n = 13)	M = 58.92 (SD = 13.20)	M = 67.77 (SD = 12.36)*	M = 59.69(SD = 11.36) ³
High mindfulness (n = 35)		M = 62.40 (SD = 12.79)	M = 70.37 (SD = 12.60)*	M = 68.54 (SD = 12.39) [@]
Low mindfulness (n = 38)		M = 57.24 (SD = 11.67)	M = 66.66 (SD = 11.01)*	M = 58.08 (SD = 10.35)
Internal imagery (n = 26)		M = 61.34 (SD = 11.41)	M = 71.77 (SD = 11.18)*	M = 67.69 (SD = 12.35) [@]
External imagery (n = 23)		M = 56.96 (SD = 14.27)	M = 64.17 (SD = 19.23)	M = 58.52 (SD = 12.82)
Control group (n = 24)		M = 60.63 (SD = 11.60)	M = 68.92 (SD = 10.66)*	M = 62.50 (SD = 10.85)

Note:

¹* $p < .05$, compared to pretest;

²# $p < .05$, compared to posttest;

³@ $p < .05$, between group comparisons;

⁴ upper part: 3-way interaction of dispositional mindfulness, imagery training and measurement time;

⁵ middle part: 2-way interaction of dispositional mindfulness and measurement time;

⁶ lower part: 2-way interaction of imagery intervention and measurement time.

2.3. Three-way interaction of dispositional mindfulness, imagery, and measurement time

A three-way 3 (treatments: internal-, external imagery, and control) X 2 (high/low mindfulness) X 3 (measurement time: pretest, posttest, and retention) mixed ANOVA statistical analysis found dispositional mindfulness interacted with treatments and measurement time, $F(4, 134) = 2.96$, $p = .022$, partial $\eta^2 = 0.081$ (Table 1). As figure 3 illustrates, when using internal imagery, high dispositional mindfulness performed better than low mindfulness in retention ($M = 74.69$ vs. $M = 60.69$), but not at the pretest ($M = 64.23$ vs. $M = 58.38$) and posttest ($M = 75.54$ vs. $M = 68.00$), $F(2, 134) = 11.06$, $p < .001$, partial $\eta^2 = 0.142$. Further, when using external imagery, high dispositional mindfulness performed better than low mindfulness in retention ($M = 64.00$ vs. $M = 53.50$) but not pretest ($M = 60.00$ vs. $M = 54.17$) and posttest ($M = 64.36$ vs. $M = 64.00$), $F(2, 134) = 13.62$, $p < .001$, partial $\eta^2 = 0.169$ (Figure 4).

2.4. Two-way interaction of dispositional mindfulness and measurement time

A two-way 2 (high/low mindfulness) X 3 (measurement time: pretest, posttest, and retention) mixed design ANOVA showed a significant interaction, $F(2, 134) = 20.57$, $p < .001$, partial $\eta^2 = 0.235$ (Figure 5). The Post hoc comparisons showed that high dispositional mindfulness performed better in retention ($M = 68.54$ vs. $M = 58.08$) than low mindfulness, $F(2, 201) = 13.98$, $p < .001$, partial $\eta^2 = 0.065$. No significant difference in the pretest ($M = 62.40$ vs. $M = 57.24$) and posttest ($M = 70.37$ vs. $M = 66.66$) between high vs. low dispositional mindfulness (see figure 5 and table 1).

2.5. Two-way interaction of imagery and measurement time

A two-way 3 (treatments: internal-, external imagery, and control) X 3 (measurement time: pretest, posttest, and retention) mixed ANOVA statistical analysis found imagery interacted with measurement time, $F(4, 134) = 46.19$, $p = .003$, partial $\eta^2 = 0.113$ (see Table 1 and Figure 6). The main effect test showed a significant difference between internal imagery and external imagery $F(2, 201) = 3.62$, $p = .028$, partial $\eta^2 = 0.035$. Specifically, internal imagery performed better than external imagery in retention ($M = 67.69$ vs. $M = 58.52$) but not pretest ($M = 61.34$ vs. $M = 56.96$) and posttest ($M = 67.69$ vs. $M = 58.52$). The control group did not differ significantly from either internal or external imagery in the pretest ($M^i = 61.34$ vs. $M^e = 56.96$ vs. $M^c = 60.63$), posttest ($M^i = 71.77$ vs. $M^e = 64.17$ vs. $M^c = 68.92$), and retention ($M^i = 67.69$ vs. $M^e = 58.52$ vs. $M^c = 62.50$).

2.6. Two-way interaction of dispositional mindfulness and imagery

Finally, we used a two-way 2 (high/low mindfulness) X 3 (treatments: internal-, external imagery, and control) mixed design ANOVA

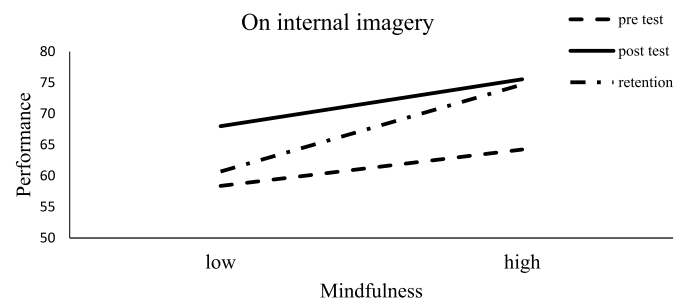


Fig. 3. Interaction of mindfulness disposition, internal imagery, and measurement time on basketball mid-range shot.

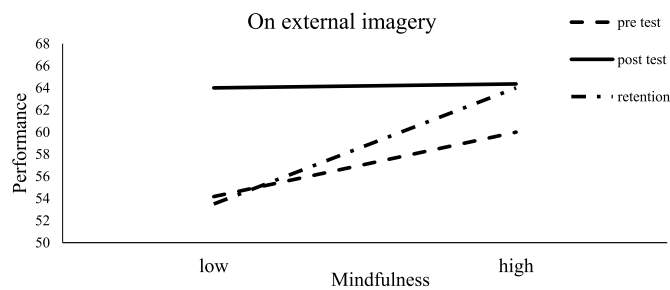


Fig. 4. Interaction of mindfulness disposition, external imagery, and measurement time on basketball mid-range shot.

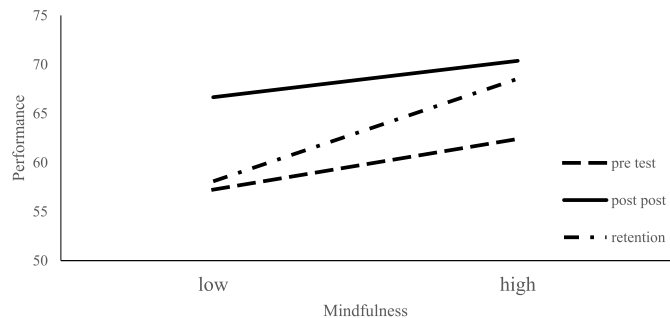


Fig. 5. Interaction of mindfulness disposition and measurement time on basketball mid-range shot.

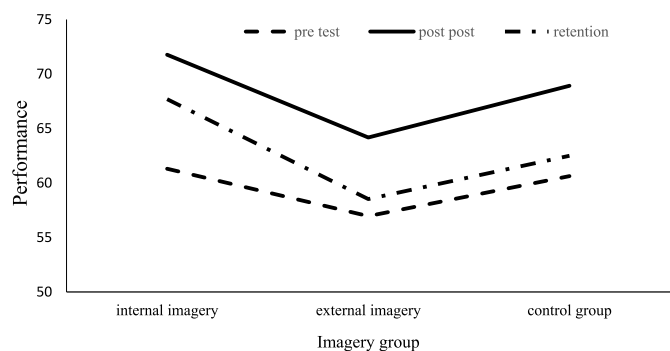


Fig. 6. Interaction of imagery intervention and measurement time on basketball mid-range shot.

and found no significant interaction $F(2, 67) = 0.32, p = .724$, partial $\eta^2 = 0.010$ (Table 1).

3. Discussion

Following past research that examined the combined effects of dispositional mindfulness and psychological intervention on motor performance, we examined whether dispositional mindfulness interacts with visualized PETTLEP imagery on basketball mid-range performance and retention. Results indicated dispositional mindfulness interacted with visualized PETTLEP imagery over different measurement times on basketball mid-range shooting. Specifically, either using internal or external imagery, high dispositional mindfulness performed better than low dispositional mindfulness in retention. Further, we found visualized PETTLEP imagery interacted with measurement time on basketball mid-range retention. The initial findings provide several implications for researchers.

3.1. Theoretical implications and explanations

First, the effects of dispositional mindfulness on basketball mid-range shooting not only support earlier studies with fine-line tracking (Chiu et al., 2019) and two-hand coordination (Liu et al., 2022) but also extend to basketball mid-range shooting. The results reflect the role of mindfulness in motor skill learning and performance. According to Birrer et al. (2012), there are 9 potential mechanisms of the impact that influence the relationship between mindfulness and personal skills and development. Among these mediators, attention, non-attachment, self-regulation, and less rumination may explain why high dispositional mindfulness performed better than low dispositional mindfulness. For attention, high dispositional mindfulness participants are possibly less distracted, better able to control their attention, and ignore irrelevant information. Thus, they performed better than their low dispositional mindfulness counterparts. For non-attachment, high dispositional mindfulness tends to reduce irrelevant mental processes and reinvestment. It is suggested that when performers try to control automatic movement consciously motor performance is impaired (Masters & Maxwell, 2008). Because of less rumination, the high dispositional mindfulness participants might have the advantage to perform basketball shooting automatically. For self-regulation, it is suggested that too high or too low arousal during movement execution will impair performance (Doyle et al., 2016). The high dispositional mindfulness tends to be focused on the present with a non-judgment attitude, this behavioral tendency might help them to perform basketball shooting in an optimal emotional state. For less rumination, it is suggested that during movement execution too much cognitive activity is detrimental to performance. It is contended that high dispositional mindfulness can reduce rumination (Raes & Williams, 2010), so it might help high-mindfulness participants to perform basketball shooting automatically.

Further, our finding that internal PETTLEP imagery performed better than external imagery in retention supports previous research that internal imagery is beneficial to closed motor skills (Hardy & Callow, 1999; McLean & Richardson, 1994; Olsson et al., 2008; Spittle, 2001). Internal imagery is defined as "... the person actually imagines being inside his/her body and experiencing those sensations which might be expected in the actual situation" (Mahoney & Avenier, 1977, p.137). Such mental processes meet the requirements of closed motor skills performed under a stationary environment that emphasizes kinesthetic experience (Gentile, 2000).

The 3-way interaction among dispositional mindfulness, the visualized PETTLEP imagery, and measurement time was a novel finding. Specifically, after the 6-week intervention, there was no posttest difference in basketball mid-range shooting among internal-, external, and control groups. However, when comparing group differences in retention, high dispositional mindfulness performed better than low dispositional mindfulness using either internal or external imagery. Our results extend earlier studies that combined dispositional mindfulness and psychological interventions (e.g., Chiu et al., 2019; Liu et al., 2022) limited to post-intervention effects. This novel finding advances knowledge of dispositional mindfulness and its function on motor skill learning and performance. The moderating effects of dispositional mindfulness in the imagery training-basketball performance relationship outline the significant contribution of dispositional mindfulness. The unique attributes of dispositional mindfulness such as pure attention, non-attachment, self-regulation, and less rumination (Birrer et al., 2012) not only enable high dispositional mindfulness to perform better than low dispositional mindfulness but also gain better training effects from psychological skill intervention.

The interactive approach to examining dispositional mindfulness and PETTLEP imagery training on sports performance offers insights for future study. According to Lewin's (1951) interaction perspective, an individual's behavior is determined by person and environment. Past imagery research generally emphasizes how different imagery training improves sports performance (e.g., Buck et al., 2016; Norouzi et al.,

2019). This approach views the environment (i.e., arrangement of imagery training) as the main source of performance variation. In contrast, emphasis on dispositional mindfulness suggests a stable trait/disposition as the main source of behavior variations (e.g., Conversano et al., 2020). Either solely emphasizing on person or environment has limitations to understanding complex human behavior. We suggest future studies that examine imagery training and sports performance include personality factors such as perfectionism (e.g., Lizmore et al., 2019), reinvestment disposition (e.g., Sparks et al., 2021), and trait anxiety (e.g., Kaur & Shenoy, 2019) into experimental design. Doing so provides a holistic picture of how imagery training affects sports performance.

There are several strengths in our study. First, we used a randomized controlled trial (RCT) to conduct our study. It is suggested that using RCT design can minimize confounding factors, selection, and assessment bias, and gain statistical reliability (Guerrera et al., 2017; Levin, 2007). So, we followed Guerrera et al. (2017) suggestions by applying a standardized study protocol, providing a clear hypothesis, and using appropriate selection criteria, sample size, and procedures for the randomization. Using a RCT experimental design elevates the level of evidence of research results (Murad et al., 2016). We suggest future studies that combine dispositional mindfulness with psychological intervention adopt the same research design to enhance validity. Further, most psychological interventions in sports focus on the immediate effects after psychological skills training (PST). It is important to examine how long a PST intervention (e.g., imagery, self-talk, or goal setting) can maintain its effect on performance without further training (i.e., retention). Retention has theoretical significance in motor skill learning and performance. It not only represents how a participant improves his/her motor performance after a psychological intervention but also indicates how long such behavioral change lasts (Lee et al., 2019).

Moreover, we adopted a visualized PETTLEP imagery approach to examine its effects on basketball mid-range shooting has several advantages for imagery training itself. According to Lu et al. (2020), the visualized imagery approach reduces the "imagery gap" between imagery scripts and real images in the imagers' mind. Traditional imagery training generally asks imagers to read or hear an imagery script to produce a real image in mind. According to Kosslyn (2005), after reading or hearing an imagery script, the imagers must recognize, interpret, and convert the imagery script into motor task images. Imagers who have hearing problems or dyslexia would have difficulty. We suggest future PETTLEP imagery studies adopt a visualized imagery approach to produce an effective intervention.

3.2. Limitations

There are several limitations to this study. First, because of the time limit, we offered a 6-week and 3-times/per week intervention; whether a greater amount of practice leads to more improvement in performance needs further examination (Schmidt et al., 2019). Second, the basketball mid-range shooting is a closed motor task; whether our results can be generalized to open motor tasks such as volleyball spiking, boxing dodging, or hockey shooting needs future study. Further, our participants are college students; whether our results can be generalized to professional basketball players or younger basketball players needs further examination. Furthermore, although participants adapted all PETTLEP elements into their learning progress, whether three times a week for six weeks are long enough to consolidate shooting stimulus-response needs further examination. Moreover, mindfulness disposition is beneficial to motor performance. However, we didn't cultivate this attribute through a mindfulness program. We suggest future studies may integrate mindfulness intervention and imagery training to examine its effects on motor performance.

3.3. Future suggestions

As previously mentioned, there are several strengths of this study

that can be useful recommendations for future research. First, this study adopted an interactive approach by combining dispositional mindfulness and imagery training to examine its effects on motor performance. A future study might consider adopting the same approach by combining performers' unique attributes (e.g., dispositional mindfulness, reinvestment disposition) and psychological skill training (e.g., mindfulness intervention, imagery) to examine its effects on motor performance. Further, this study adopted a RCT research design. A future study might adopt the same approach to minimize selection and assessment bias, and gain statistical reliability (Guerrera et al., 2017; Levin, 2007).

3.4. Applications

To apply our results in PE or sports settings, we suggest PE teachers or sports coaches use participants' own successful sports videos to do imagery training. Participants' own successful video not only builds confidence but also increases motivation to engage in the intervention (Lu et al., 2020). Further, because past research found a brief mindfulness intervention will bring performers into a state of mindfulness and concentration (e.g., Kee et al., 2012), we suggest PE teachers or sports coaches use a similar technique before sports performance. By doing so, it is not only enhancing their mental state and performance but also cultivating their dispositional mindfulness.

4. Conclusion

By adopting Lewin's (1951) interactive approach and RCT design, we found the moderating role of dispositional mindfulness in the psychological skill training-skill performance relationship. The new finding inspires researchers' thoughts about the role of dispositional mindfulness, research design, and psychological training. Further, to enhance performers' motor skill performance, we suggested PE teachers or sports coaches can teach a brief mindfulness intervention before sports performance or cultivate dispositional mindfulness through daily practice.

Declaration of conflicting interests

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Data availability

Data will be made available on request.

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